

SET-PARTITION $\leq_p$ RECTANGLE PACKING

set-partition problem is, given

a set of integers $S = \{a_1, a_2, \dots, a_n\}$,

find whether there exists a subset

$$S' \subseteq S \text{ s.t. } \sum_{x \in S'} x = \sum_{x \in S \setminus S'} x$$

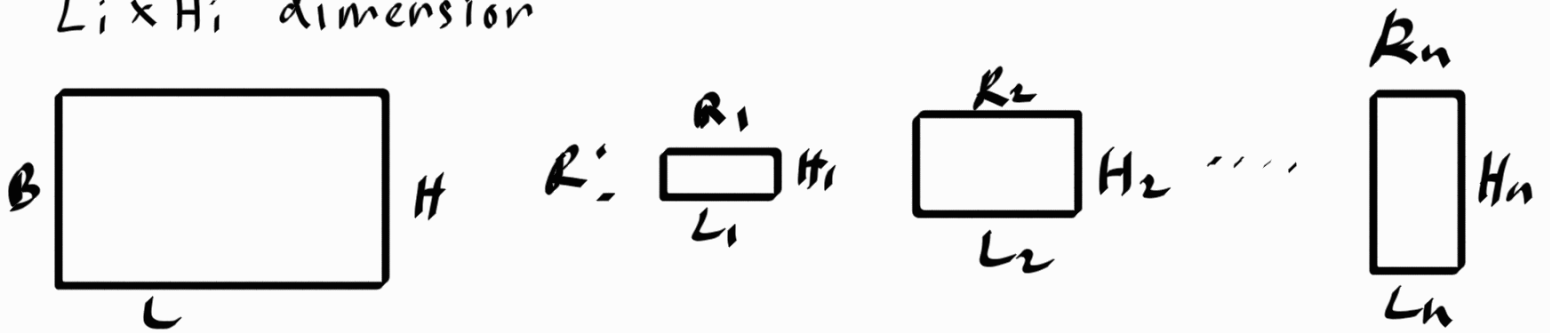
$$S = \{8, 9, 11, 2, 3, 5\}$$

$$\{8, 11\} \quad \text{and} \quad \{9, 2, 3, 5\}$$

Rectangle Packing

Given a container B of size $L \times H$,

You are given a set of Rectangles R ,
 $R = \{R_1, R_2, R_3, \dots, R_n\}$, where R_i is of
 $L_i \times H_i$ dimension



Reduction:

Let the input to the set partition be

$$S = \{a_1, a_2, a_3, \dots, a_n\}$$

$\forall a_i$, construct R_i with length a_i and height ϵ

- B is constructed to have length $\frac{\sum a_i}{2}$ and height 2ϵ

* There is a solution for set partition instance if and only if all R_i 's can be packed into B .

$$S = \{8, 9, 11, 2, 3, 5\}$$
$$\{8, 11\} \quad \{9, 2, 3, 5\}$$

8	11				}ε
9	1	2	3	5	

$$19 = \frac{\sum a_i}{2}$$

S has set partition solution $\Leftrightarrow R$ can be packed into B

Reduction takes $O(n)$ time

What is NP-complete?

Recap

A problem B is NP-complete, If

1. $B \in NP$

2. Every Problem in NP can be polynomial time reducible to B.

SAT is the first NP-complete problem that was proved using the above definition directly.

we say B is NP-complete

1. $B \in NP$

2. $C \leq_p B$, where C is NP-complete

Suppose A is any NP problem

$A \leq_p C \leq_p B$

What is NP-hard?

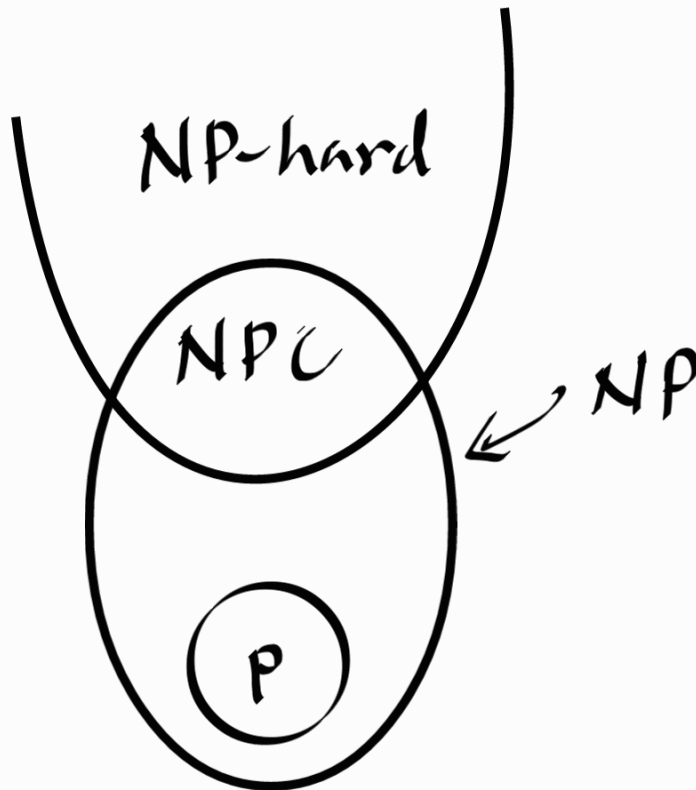
Suppose C is NP-complete.

Also suppose $C \leq_p A$

If we do not show that A is in NP,
then we cannot say A is NPC.

But, we can say A is NP-hard.

Ex: optimization problems of
NP-complete problems are NP-hard.



SAT (satisfiability problem) is NP-complete.

$$\text{Input } \phi = (x_1 \vee x_2 \vee \bar{x}_3 \vee x_4) \wedge (x_2 \vee \bar{x}_3 \vee x_4 \vee \bar{x}_6) \\ \wedge (x_7 \vee \bar{x}_1) \dots$$

Q: Is there a truth assignment that satisfies the ϕ

3-SAT: A special case of SAT problem, where every clause has 3 literals.

$$\text{Ex: } \phi = (x_1 \vee x_2 \vee \bar{x}_3) \wedge (x_4 \vee \bar{x}_5 \vee x_6) \wedge \dots \\ \dots \wedge (x_2 \vee \bar{x}_3 \vee \bar{x}_5)$$

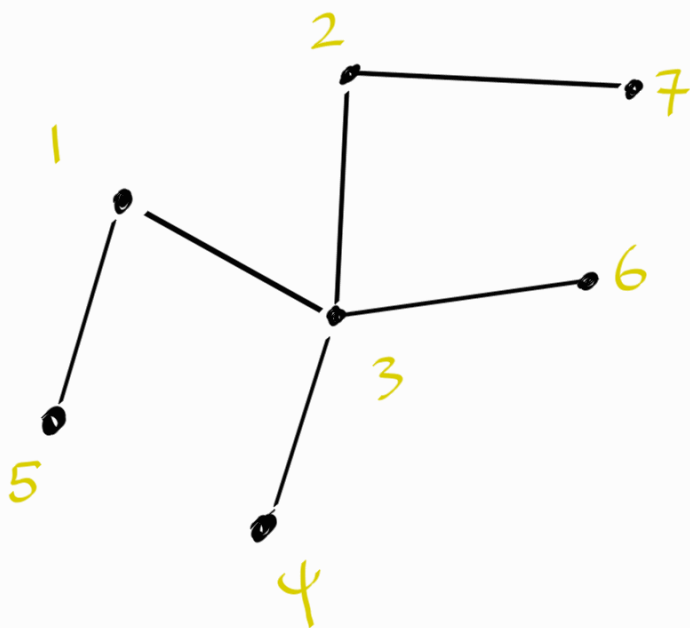
Let's look at how we can show a problem is NP-complete

we want to show the decision version of vertex-cover problem is NP-complete.

what is min-vertex-cover problem?

Given a graph $G = (V, E)$, find the minimum number of vertices that covers all edges.

Ex:



$\{1, 3, 7\}$

This is an optimization problem.

Let's convert this to a decision problem.

Given graph $G=(V,E)$, and $k \in \mathbb{N}$,
is there a vertex cover of size at
most k ?

Definition of NP-completeness

1. The problem is in NP
2. Every problem in NP is
Polynomial time reducible to
the problem considering

Proof: Let's prove $VC \in NP$.

Sketch:

1. $VC \in NP$
2. $3SAT \leq_P VC$

Let's create an NTM that decides VC .

$N =$ on input $\langle G, k \rangle$

1. Nondeterministically select k nodes from G .
2. Delete the k nodes and edges incident to it. If there are edges left, reject; otherwise accept.

Clearly deletion of the nodes and edges can be done in $O(n^2)$

Second we need to show
any problem is NP can be
polynomially reducible to VC.

$$3SAT \leq_p VC$$

Proof: Suppose ϕ has m variables
and l clauses

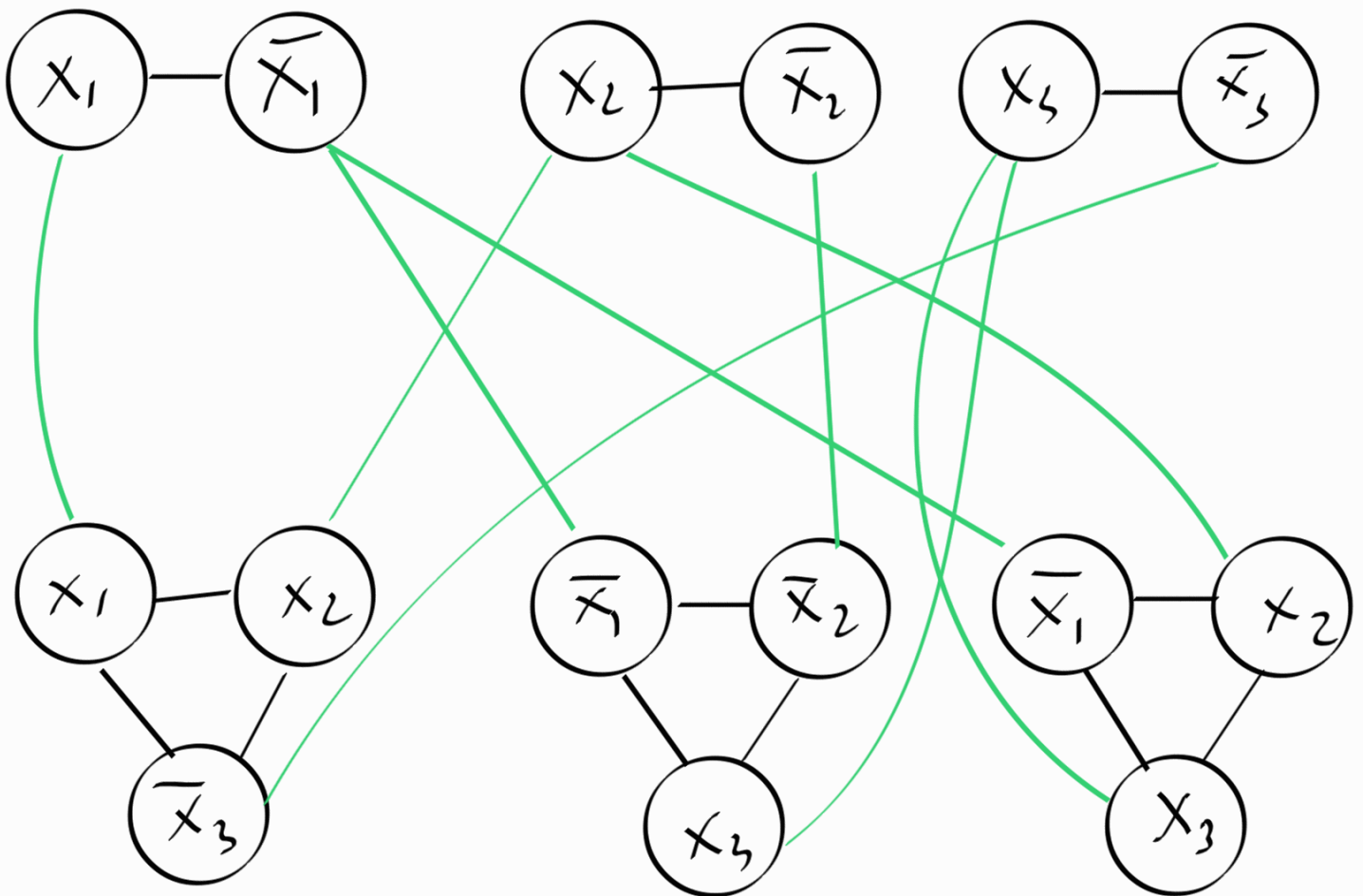
$$\text{Ex! } \phi = (x_1 \vee x_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee x_3) \wedge (x_1 \vee x_2 \vee x_3)$$

m -variables
 l -clauses,

Truth assignment?

$$x_1 = T, x_2 = F, x_3 = T$$

variable gadget



Clause gadgets

The size of the graph G ?

— ϕ has a truth assignment
iff G has a VC of size
 $k = m + 2l$ vertices.

(i)

$\Rightarrow$ If ϕ has a truth assignment
 G has a VC of size $k = m + 2l$

(ii)

$\Leftarrow$ If G has a VC of size
 $m + 2l$, then ϕ has a truth
assignment.